

Loading Historical Transactions Data into NoSQL Database

Commands to load the past transactions data into NoSQL database

Create a file name as **createHistorical.hql** and add below SQL commands to the file.

1. Create **card_transactions_stg** staging table to load raw csv into staging table.

```
CREATE EXTERNAL TABLE IF NOT EXISTS card_transactions_stg (  
  card_id STRING,  
  member_id STRING,  
  amount DOUBLE,  
  postcode STRING,  
  pos_id STRING,  
  transaction_dt STRING,  
  status STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
tblproperties("skip.header.line.count"="1");
```

2. Load the data from csv file from hadoop env to card_transactions_stg table.

```
LOAD DATA INPATH 'hdfs:/user/hadoop/hive/card_transactions.csv'  
OVERWRITE INTO TABLE card_transactions_stg;
```

3. Create **card_transactions** table and store the table in Hbase system.

```
CREATE EXTERNAL TABLE IF NOT EXISTS card_transactions (row_key  
  struct<card_id:string, pos_id:string, transaction_dt:string,  
  amount:DOUBLE>, card_id STRING, pos_id STRING, transaction_dt STRING,  
  member_id STRING, amount DOUBLE, postcode STRING, status STRING)  
ROW FORMAT DELIMITED  
COLLECTION ITEMS TERMINATED BY '~'  
STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'
```

```
WITH SERDEPROPERTIES ('hbase.columns.mapping'='TD:card_id,
TD:pos_id, TD:transaction_dt,TD:member_id,TD:amount, TD:postcode,
TD:status')
TBLPROPERTIES ("hbase.table.name" = "card_transactions");
```

4. Insert data into **card_transactions** table from staging table to make it compatible with hbase.

```
Insert into card_transactions select
NAMED_STRUCT('card_id',card_id,'pos_id',pos_id,'transaction_dt',
transaction_dt) as row_key,card_id,pos_id,transaction_dt, member_id,
amount, postcode, status from card_transactions_stg;
```

5. Run file historical.hql using below command:
hive -f createHistorical.hql
6. Finally create a wrapper shell script to copy the **card_transactions.csv** file to hive environment, create the necessary tables in Hbase & run the **createHistorical.hql** script in hive.

```
[hadoop@ip-172-31-5-80 ~]$ vi loadNosql.sh
[hadoop@ip-172-31-5-80 ~]$ cat loadNosql.sh
#!/bin/bash

hdfs dfs -rm -r hdfs://user/hadoop/hive
hdfs dfs -mkdir hive
hdfs dfs -copyFromLocal /home/hadoop/card_transactions.csv /user/hadoop/hive/card_transactions.csv

echo "create 'card_transactions', 'TD'" | sudo hbase shell -n
echo "create 'look_up_table', 'card_details', 'Member_details', 'Location', 'Rule_params'" | sudo hbase shell -n

hive -f /home/hadoop/createHistorical.hql
[hadoop@ip-172-31-5-80 ~]$
```

Screenshot of the steps performed to create the tables & validating counts in Hive & Hbase.

```
[hadoop@ip-172-31-5-80 ~]$ cat createHistorical.hql
CREATE EXTERNAL TABLE IF NOT EXISTS card_transactions_stg (
  card_id STRING,
  member_id STRING,
  amount DOUBLE,
  postcode STRING,
  pos_id STRING,
  transaction_dt STRING,
  status STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
LOCATION '/user/hadoop/historical/card_transactions'
tblproperties("skip.header.line.count"="1");

LOAD DATA INPATH 'hdfs://user/hadoop/hive/card_transactions.csv' OVERWRITE INTO TABLE card_transactions_stg;

CREATE EXTERNAL TABLE IF NOT EXISTS card_transactions (row_key struct<card_id:string, pos_id:string, transaction_dt:string>, card_id STRING, pos_id STRING, transaction_dt STRING, member_id STRING,amount DOUBLE, postcode STRING,status STR
ING)
ROW FORMAT DELIMITED
COLLECTION ITEMS TERMINATED BY '-'
STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'
WITH SERDEPROPERTIES ('hbase.columns.mapping'='TD:card_id, TD:pos_id, TD:transaction_dt,TD:member_id,TD:amount, TD:postcode, TD:status')
TBLPROPERTIES ("hbase.table.name" = "card_transactions");

Insert into card_transactions select NAMED_STRUCT('card_id',card_id,'pos_id',pos_id,'transaction_dt', transaction_dt) as row_key,card_id,pos_id,transaction_dt, member_id, amount, postcode, status from card_transactions_stg;

CREATE VIEW IF NOT EXISTS last_ten_transactions
AS select card_id, member_id, amount, transaction_dt, postcode, rank() over (PARTITION BY card_id ORDER BY unix_timestamp(transaction_dt , 'dd-MM-yyyy hh:mm:ss') desc, amount desc) as ranking from card_transactions where status='GENUINE'
;
```

```
[hadoop@ip-172-31-5-80 ~]$ ls -lrt
total 8712
drwxr-xr-x 3 hadoop hadoop    149 Apr 27  2021 mysql-connector-java-8.0.25
-rw-rw-r-- 1 hadoop hadoop 4079310 Aug  7  2021 mysql-connector-java-8.0.25.tar.gz
-rw-rw-r-- 1 hadoop hadoop 4829520 Jan 17 07:48 card_transactions.csv
-rw-rw-r-- 1 hadoop hadoop   392 Jan 19 13:34 loadNosql.sh
-rw-rw-r-- 1 hadoop hadoop  1520 Jan 19 13:40 createHistorical.hql
[hadoop@ip-172-31-5-80 ~]$

[hadoop@ip-172-31-5-80 ~]$ sh loadNosql.sh
rm: 'hdfs://user/hadoop/hive': No such file or directory
0 row(s) in 2.6260 seconds

Hbase::Table - card_transactions
0 row(s) in 2.6100 seconds

Hbase::Table - look_up_table

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
OK
Time taken: 1.423 seconds
Loading data to table default.card_transactions_stg
OK
Time taken: 2.166 seconds
OK
Time taken: 1.293 seconds
Query ID = hadoop_20240119134254_4b355a4a-da7f-497b-a97f-e3f0526f6c2f
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1705670818466_0002)

-----
VERTICES    MODE           STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container    SUCCEEDED    1         1         0         0         0         0
-----
VERTICES: 01/01  [=====] 100% ELAPSED TIME: 6.79 s
-----
OK
Time taken: 31.788 seconds
OK
Time taken: 0.343 seconds
[hadoop@ip-172-31-5-80 ~]$
```

Hbase Table:

card_transactions table created with 53292 records (Screenshot below).

Command to list the table in which the data is loaded and the command to get the count of the rows of the table.

Hbase command:

```
list
count 'card_transactions'
```

```
=> ["card_transactions", "look_up_table"]
hbase(main):002:0> count 'card_transactions'
Current count: 1000, row: 341719092861087~348479157173653~09-12-2017 01:50:04~5777101.0
Current count: 2000, row: 343252340741735~878481175923264~17-03-2016 01:27:55~499306.0
Current count: 3000, row: 344453305077666~477889498690453~03-05-2017 13:17:03~4112459.0
Current count: 4000, row: 345864096406128~263597298380498~10-01-2018 13:19:13~5620846.0
Current count: 5000, row: 347177017703006~607111050994623~24-11-2017 19:29:35~3881266.0
Current count: 6000, row: 348390084708787~252122735905449~31-08-2017 15:58:36~4365570.0
Current count: 7000, row: 349261557797449~220369360742912~10-11-2017 13:09:04~4130551.0
Current count: 8000, row: 371085417506954~790236765996611~22-08-2016 15:36:09~3667024.0
Current count: 9000, row: 372838963275954~191667371054185~06-04-2017 08:12:05~5693002.0
Current count: 10000, row: 374437449333250~104804337714838~12-11-2016 17:23:01~9084333.0
Current count: 11000, row: 375956299108731~733657582169915~10-08-2017 23:08:32~9766446.0
Current count: 12000, row: 377037030364630~605988874514427~06-10-2017 12:13:10~2332982.0
Current count: 13000, row: 378118026791597~586336168666572~25-08-2017 19:31:43~1280928.0
Current count: 14000, row: 4009218272111551~298935629685938~01-08-2016 22:46:05~8656880.0
Current count: 15000, row: 4067184730430984~275178801561140~21-02-2017 14:21:14~6830136.0
Current count: 16000, row: 4132381122041426~598314604248219~10-12-2016 21:17:21~9303987.0
Current count: 17000, row: 4264553579186587~858567976578612~18-04-2017 23:55:12~6058344.0
Current count: 18000, row: 4373464339970856~657196939516344~11-01-2018 00:00:00~6559716.0
Current count: 19000, row: 4439501833420177~634293040387939~17-02-2017 05:01:05~5371581.0
Current count: 20000, row: 4500499651579063~727121920702488~08-12-2017 13:32:44~3513696.0
Current count: 21000, row: 4566853351752365~335349979924073~03-04-2017 20:30:40~3669382.0
Current count: 22000, row: 4600996294769125~259895316688347~25-12-2017 00:33:18~7070746.0
Current count: 23000, row: 4689314809377828~989600253556389~24-10-2016 02:52:59~185332.0
Current count: 24000, row: 4766789897935106~072938202866006~09-10-2017 20:58:38~4206082.0
Current count: 25000, row: 4863127030291206~483187165654021~12-05-2016 23:19:14~3878839.0
Current count: 26000, row: 4907253800863053~776565941881281~14-12-2017 09:59:35~7434309.0
Current count: 27000, row: 4995478705000641~155223326542744~20-05-2017 02:02:58~1780486.0
Current count: 28000, row: 5134479292018417~128069614660231~08-11-2017 08:29:00~4083279.0
Current count: 29000, row: 5162808285745682~924897469747000~02-07-2016 15:57:01~1888498.0
Current count: 30000, row: 5186615811954262~026744451387425~23-01-2018 06:47:40~2972350.0
Current count: 31000, row: 5221229593682054~127498352656607~16-08-2017 16:04:44~8870294.0
Current count: 32000, row: 5258095619226135~524892536444302~11-11-2017 00:00:00~5577468.0
Current count: 33000, row: 5294592751808411~698043686524445~28-12-2017 19:50:46~792320.0
Current count: 34000, row: 5342400571435088~569800919443173~26-08-2016 07:22:56~2049055.0
Current count: 35000, row: 5380978184175608~667233655872426~01-09-2016 10:51:09~9057468.0
Current count: 36000, row: 5414439899219272~345244546468970~07-06-2017 00:10:15~1413268.0
Current count: 37000, row: 5481808794715436~234167732114687~20-10-2017 16:48:20~3012181.0
Current count: 38000, row: 5534323829711423~637125777249291~30-06-2017 11:43:05~535263.0
Current count: 39000, row: 5584977018799504~770952296643375~03-05-2017 09:46:23~9635501.0
Current count: 40000, row: 6011139413319542~936287997923127~30-03-2016 07:13:43~1908818.0
Current count: 41000, row: 6011525010455848~959531034098755~02-08-2017 23:46:55~2762867.0
Current count: 42000, row: 6011782857327719~167569302823461~09-10-2016 02:59:08~8187310.0
Current count: 43000, row: 6221796595498984~174952353920145~18-02-2016 01:04:16~7689490.0
Current count: 44000, row: 6224271253849917~048019406374284~12-12-2017 19:22:13~4960131.0
Current count: 45000, row: 6225606551069826~648178895653883~13-07-2017 13:58:12~4112360.0
Current count: 46000, row: 6228733641419063~422091580968713~11-05-2016 10:27:11~2641410.0
Current count: 47000, row: 6447877814927926~992747968210744~11-01-2018 00:00:00~9774152.0
Current count: 48000, row: 6461356425954109~839304530643246~03-11-2016 20:57:23~1877568.0
Current count: 49000, row: 6480152634975473~963893207999520~17-01-2018 17:22:08~5863997.0
Current count: 50000, row: 6505080237250161~874636482279942~18-09-2017 17:36:33~2156682.0
Current count: 51000, row: 6544876671165176~424648847023528~04-08-2017 16:09:39~5155240.0
Current count: 52000, row: 6574255180086418~504711588075355~20-12-2017 08:11:48~5899422.0
Current count: 53000, row: 6595814135833988~311764663134170~18-11-2017 12:30:42~2717739.0
53292 row(s) in 2.2530 seconds

=> 53292
hbase(main):003:0> exit()
(hadoop@ip-172-31-5-175 ~)$
```

Hive Table:

card_transactions and **card_transactions_stg** Table created with 53292 records.

Hive Command:

```
select count(*) from card_transactions;
```

```
select count(*) from card_transactions_stg;
```

```
[hadoop@ip-172-31-5-175 ~]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
hive> show tables;
OK
card_transactions
card_transactions_stg
last_ten_transactions
look_up_table
Time taken: 0.526 seconds, Fetched: 4 row(s)
hive> select count(*) from card_transactions;
Query ID = hadoop_20240117154216_87fbf425-ee48-4725-a348-1d670fabadbe
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1705496453063_0012)

-----
      VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Reducer 2 ..... container  SUCCEEDED    1         1         0         0         0         0
-----
VERTICES: 02/02  [=====>>] 100%  ELAPSED TIME: 6.69 s
-----
OK
53292
Time taken: 12.49 seconds, Fetched: 1 row(s)
hive>

hive> select count(*) from card_transactions_stg;
Query ID = hadoop_20240117145456_2cc09865-d627-4b20-854a-1f048a207ade
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1705496453063_0005)

-----
      VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Reducer 2 ..... container  SUCCEEDED    1         1         0         0         0         0
-----
VERTICES: 02/02  [=====>>] 100%  ELAPSED TIME: 4.43 s
-----
OK
53292
Time taken: 4.977 seconds, Fetched: 1 row(s)
hive>
```